

Learning math concepts as program induction

June 26, 2026

1 Formalism for arithmetic reasoning

We formalize arithmetic reasoning as a sequential decision process over symbolic expressions. Expressions are represented as **FlatDAGs**, flat, sequential structures that capture the atoms and operators of an arithmetic expression at each level of nesting. A learner’s reasoning step is a local rewrite: given the current FlatDAG state, they select an operator and reduce it, producing a new FlatDAG. The set of available rewrites at each step is determined by pattern matching over 3-node windows in the current expression.

1.1 Expression Representation

We represent arithmetic expressions as **FlatDAGs** labeled, ordered, directed acyclic graphs whose structure reflects the flat, sequential nature of arithmetic expressions at a single level of nesting.

Formally, a FlatDAG is a pair $(\mathbf{a}, \mathbf{op})$ where:

- $\mathbf{a} = [a_0, a_1, \dots, a_n]$ is an ordered list of **atoms**. Each atom is either a numeric literal $q \in \mathbb{Q}_{\geq 0}$, a unary node $u \in \{u-, u+\}$ applied to an atom, or a bracket node $(\cdot)$ whose interior is itself a FlatDAG recursively.
- $\mathbf{op} = [op_0, op_1, \dots, op_{n-1}]$ is an ordered list of **operator nodes**, where each $op_i \in \{+, -, \times, \div\}$ sits between a_i and a_{i+1} , with $op_i.\text{left} = a_i$ and $op_i.\text{right} = a_{i+1}$.

The shared-object edges, where $op_i.\text{right} = a_{i+1} = op_{i+1}.\text{left}$ are what make this structure a DAG rather than a tree: adjacent operators share atom nodes.

Example 1. *Given the expression $5 + 4 \times (6 - 9)$, the FlatDAG at the top level is:*

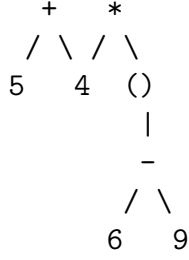
$$\mathbf{a} = [5, 4, (6 - 9)], \quad \mathbf{op} = [+ , \times]$$

where $(6 - 9)$ is a bracket node whose interior is itself a FlatDAG:

$$\mathbf{a}' = [6, 9], \quad \mathbf{op}' = [-]$$

A FlatDAG is constructed from an expression string via a recursive parser that tokenizes the input, alternates between parsing atoms and operators at each level, and wraps any parenthesized subexpression as a bracket node containing its own inner FlatDAG.

For example, the representation of $5 + 4 * (6 - 9)$ would be:



1.2 Valid Action Identification

The patterns here, are from the tables below (Tables 1 - 5)

Given a FlatDAG with operator list $\mathbf{op} = [op_0, op_1, \dots, op_{n-1}]$, we identify which operators a correct learner may evaluate by intersecting the constraints imposed by all 3-node windows the operator participates in.

Each operator op_i participates in at most two windows as the left operator of the window $(a_{i-1}, op_{i-1}, a_i, op_i, a_{i+1})$ and as the right operator of the window $(a_i, op_i, a_{i+1}, op_{i+1}, a_{i+2})$. An operator op_i is **valid** if and only if every window it participates in marks it as the correct move.

Consider $4 + 5 + 6 \times 2 + 3$ with $\mathbf{op} = [+_1, +_2, \times_3, +_4]$, matching patterns:

- $P_1 : a + b + c$,
- $P_2 : a + b \times c$
- $P_3 : a \times b + c$

	P_1	P_2	P_3	$P_1 \wedge P_2 \wedge P_3$
$+_1$	T	—	—	T
$+_2$	T	F	—	F
$\times_3$	—	T	T	T
$+_4$	—	—	F	F

So the valid operators are $+_1$ and $\times_3$, giving two possible next steps: $9 + 6 \times 2 + 3$ or $4 + 5 + 12 + 3$.

So for any expression, the valid actions are given by -

$$\text{ValidActions}(G) = \left\{ op_i \left| \bigwedge_{P \ni op_i} P(op_i) = \top \right. \right\}$$

where $P \ni op_i$ denotes all 3-node window patterns that op_i participates in, and $P(op_i) = \top$ means pattern P marks op_i as a correct move.

To define operator precedence, We define three variables used throughout: a, b, c denote literal numbers. The tables below catalog all reduction patterns visible within a three-node sliding window. Table 1 covers the basic reduction operations and Table 2 covers the case where all three slots are literals (two reductions possible). Table 3,4 and 5 cover the reduction rules where an unreduced term Y of the form $(\dots)$ is present.

Table 1: Basic Reduction Operations

Pattern	Reduces To
$\$a + b\$$	$\llbracket a + b \rrbracket$
$\$a - b\$$	$\llbracket a - b \rrbracket$
$\$a \times b\$$	$\llbracket a \times b \rrbracket$
$\$a \div b\$$	$\llbracket a \div b \rrbracket$
$(a + b)$	$(\llbracket a + b \rrbracket)$
$(a - b)$	$(\llbracket a - b \rrbracket)$
$(a \times b)$	$(\llbracket a \times b \rrbracket)$
$(a \div b)$	$(\llbracket a \div b \rrbracket)$
(a)	a

Table 2: Three-Node Window: All Literals

Pattern	Reduces To
$a + b + c$	$\llbracket a + b \rrbracket + c$ $a + \llbracket b + c \rrbracket$
$a + b - c$	$\llbracket a + b \rrbracket - c$ $a + \llbracket b - c \rrbracket$
$a + b \times c$	$\llbracket a + b \rrbracket \times c$ $a + \llbracket b \times c \rrbracket$
$a + b \div c$	$\llbracket a + b \rrbracket \div c$ $a + \llbracket b \div c \rrbracket$
$a - b + c$	$\llbracket a - b \rrbracket + c$ $a - \llbracket b + c \rrbracket$
$a - b - c$	$\llbracket a - b \rrbracket - c$ $a - \llbracket b - c \rrbracket$
$a - b \times c$	$\llbracket a - b \rrbracket \times c$ $a - \llbracket b \times c \rrbracket$
$a - b \div c$	$\llbracket a - b \rrbracket \div c$ $a - \llbracket b \div c \rrbracket$
$a \times b + c$	$\llbracket a \times b \rrbracket + c$ $a \times \llbracket b + c \rrbracket$
$a \times b - c$	$\llbracket a \times b \rrbracket - c$ $a \times \llbracket b - c \rrbracket$
$a \times b \times c$	$\llbracket a \times b \rrbracket \times c$ $a \times \llbracket b \times c \rrbracket$
$a \times b \div c$	$\llbracket a \times b \rrbracket \div c$ $a \times \llbracket b \div c \rrbracket$
$a \div b + c$	$\llbracket a \div b \rrbracket + c$ $a \div \llbracket b + c \rrbracket$
$a \div b - c$	$\llbracket a \div b \rrbracket - c$ $a \div \llbracket b - c \rrbracket$
$a \div b \times c$	$\llbracket a \div b \rrbracket \times c$ $a \div \llbracket b \times c \rrbracket$
$a \div b \div c$	$\llbracket a \div b \rrbracket \div c$ $a \div \llbracket b \div c \rrbracket$

Table 3: Three-Node Window: Y on Right

Pattern	Reduces To
$a + b + Y$	$\llbracket a + b \rrbracket + Y$ $a + b + \llbracket Y \rrbracket$
$a + b - Y$	$\llbracket a + b \rrbracket - Y$ $a + b - \llbracket Y \rrbracket$
$a + b \times Y$	$\llbracket a + b \rrbracket \times Y$ $a + b \times \llbracket Y \rrbracket$
$a + b \div Y$	$\llbracket a + b \rrbracket \div Y$ $a + b \div \llbracket Y \rrbracket$
$a - b + Y$	$\llbracket a - b \rrbracket + Y$ $a - b + \llbracket Y \rrbracket$
$a - b - Y$	$\llbracket a - b \rrbracket - Y$ $a - b - \llbracket Y \rrbracket$
$a - b \times Y$	$\llbracket a - b \rrbracket \times Y$ $a - b \times \llbracket Y \rrbracket$
$a - b \div Y$	$\llbracket a - b \rrbracket \div Y$ $a - b \div \llbracket Y \rrbracket$
$a \times b + Y$	$\llbracket a \times b \rrbracket + Y$ $a \times b + \llbracket Y \rrbracket$
$a \times b - Y$	$\llbracket a \times b \rrbracket - Y$ $a \times b - \llbracket Y \rrbracket$
$a \times b \times Y$	$\llbracket a \times b \rrbracket \times Y$ $a \times b \times \llbracket Y \rrbracket$
$a \times b \div Y$	$\llbracket a \times b \rrbracket \div Y$ $a \times b \div \llbracket Y \rrbracket$
$a \div b + Y$	$\llbracket a \div b \rrbracket + Y$ $a \div b + \llbracket Y \rrbracket$
$a \div b - Y$	$\llbracket a \div b \rrbracket - Y$ $a \div b - \llbracket Y \rrbracket$
$a \div b \times Y$	$\llbracket a \div b \rrbracket \times Y$ $a \div b \times \llbracket Y \rrbracket$
$a \div b \div Y$	$\llbracket a \div b \rrbracket \div Y$ $a \div b \div \llbracket Y \rrbracket$

Table 4: Three-Node Window: Y on Left

Pattern	Reduces To
$Y + a + b$	$Y + \llbracket a + b \rrbracket$ $\llbracket Y \rrbracket + a + b$
$Y + a - b$	$Y + \llbracket a - b \rrbracket$ $\llbracket Y \rrbracket + a - b$
$Y + a \times b$	$Y + \llbracket a \times b \rrbracket$ $\llbracket Y \rrbracket + a \times b$
$Y + a \div b$	$Y + \llbracket a \div b \rrbracket$ $\llbracket Y \rrbracket + a \div b$
$Y - a + b$	$Y - \llbracket a + b \rrbracket$ $\llbracket Y \rrbracket - a + b$
$Y - a - b$	$Y - \llbracket a - b \rrbracket$ $\llbracket Y \rrbracket - a - b$
$Y - a \times b$	$Y - \llbracket a \times b \rrbracket$ $\llbracket Y \rrbracket - a \times b$
$Y - a \div b$	$Y - \llbracket a \div b \rrbracket$ $\llbracket Y \rrbracket - a \div b$
$Y \times a + b$	$Y \times \llbracket a + b \rrbracket$ $\llbracket Y \rrbracket \times a + b$
$Y \times a - b$	$Y \times \llbracket a - b \rrbracket$ $\llbracket Y \rrbracket \times a - b$
$Y \times a \times b$	$Y \times \llbracket a \times b \rrbracket$ $\llbracket Y \rrbracket \times a \times b$
$Y \times a \div b$	$Y \times \llbracket a \div b \rrbracket$ $\llbracket Y \rrbracket \times a \div b$
$Y \div a + b$	$Y \div \llbracket a + b \rrbracket$ $\llbracket Y \rrbracket \div a + b$
$Y \div a - b$	$Y \div \llbracket a - b \rrbracket$ $\llbracket Y \rrbracket \div a - b$
$Y \div a \times b$	$Y \div \llbracket a \times b \rrbracket$ $\llbracket Y \rrbracket \div a \times b$
$Y \div a \div b$	$Y \div \llbracket a \div b \rrbracket$ $\llbracket Y \rrbracket \div a \div b$

Table 5: Three-Node Window: Y in Middle (Recurse)

Pattern	Reduces To
$a + Y + b$	$a + \llbracket Y \rrbracket + b$
$a + Y - b$	$a + \llbracket Y \rrbracket - b$
$a + Y \times b$	$a + \llbracket Y \rrbracket \times b$
$a + Y \div b$	$a + \llbracket Y \rrbracket \div b$
$a - Y + b$	$a - \llbracket Y \rrbracket + b$
$a - Y - b$	$a - \llbracket Y \rrbracket - b$
$a - Y \times b$	$a - \llbracket Y \rrbracket \times b$
$a - Y \div b$	$a - \llbracket Y \rrbracket \div b$
$a \times Y + b$	$a \times \llbracket Y \rrbracket + b$
$a \times Y - b$	$a \times \llbracket Y \rrbracket - b$
$a \times Y \times b$	$a \times \llbracket Y \rrbracket \times b$
$a \times Y \div b$	$a \times \llbracket Y \rrbracket \div b$
$a \div Y + b$	$a \div \llbracket Y \rrbracket + b$
$a \div Y - b$	$a \div \llbracket Y \rrbracket - b$
$a \div Y \times b$	$a \div \llbracket Y \rrbracket \times b$
$a \div Y \div b$	$a \div \llbracket Y \rrbracket \div b$